OMIS 3020

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Group Project - Group 1
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March 24, 2026

Early Detection of Diabetes Risk Using Predictive Modelling

1. Introduction

Diabetes is a growing public health concern in Canada, which is driven by its aging population as well as a rise in inactive lifestyles, unhealthy diets, and obesity. As the number of people suffering from diabetes continues to grow, this has put immense pressure on Canada's healthcare system. As a consequence, diabetes has the potential to bring on serious health consequences if not detected at an early stage. However, once detected, this condition has the potential to alter the life of a patient to a considerable degree.

We chose this topic to assess how predictive analytics could be utilized to ease this healthcare burden. The objective of this project is to design an accurate predictive classifier that uses indicators in order to predict whether one has diabetes or not. We used a comprehensive dataset that comprises 17 different risk factors. Though this dataset has been collected from China, indicators such as Fasting Plasma Glucose, Body Mass Index, and blood pressure are universally accepted markers that have the potential to influence the development of diabetes. Therefore, this data has immense potential to be directly applicable to Canadian preventive healthcare.

2. Data Preprocessing

The dataset consists of 4303 rows and 18 variables which includes both numerical and categorical features. A data type check was performed using df.info() to understand more about the dataset. The results showed that most variables were stored as numerical type.There are no missing values detected across all the variables. In the dataset, smoking and drinking dataset was stored as numerical values but it represents categorical information. Therefore, further processing will be implemented to ensure data consistency and correctness.

For both smoking and drinking variables, the only valid category codes are 1,2 and 3. There were values that are outside this range and it was treated as invalid and replaced with missing values (NaN). The missing values were then imputed using the mode of each respective variable as mode imputation is the most appropriate for categorical data. After imputation, the numerical codes were mapped to their corresponding categorical labels ("Current", "Ever", "Never") to improve the interpretability.

Additionally, the column "family_histroy" was renamed to "family_history". This is to correct a typographical error and maintain the consistency in variable naming.

Outlier analysis was performed on all continuous numerical variables using boxplots. Categorical variables were excluded as outliers are not meaningful for non-numerical data. The results showed that ALT, SBP, CCR and DBP contained the most prominent outliers,

with ALT and CCR showing a strong right-skewness (Figure 1). There were also moderate outliers observed in the variables.

However, there were no outliers removed as all values are likely to represent real clinical variations rather than human errors. If the outliers were removed from the dataset, it might reduce the model's ability to capture meaningful patterns and to identify diabetes cases. The impact of the outliers will be managed through appropriate scaling and model selection.

3. Visualization

Prior to training our prediction models, we conducted Exploratory Data Analysis (EDA) to understand the underlying patterns within our dataset. To achieve this, we focused our visualization efforts on the variables that had the most significant correlation with our target variable, which is Diabetes.

To begin with, we began our baseline demographic analysis to understand that there is a significant age gap between patients and that most of our patient pool is comprised of males (~65%). Next, we created side-by-side bar charts to clearly isolate the most impactful biochemical variables. Through this type of explicit visualization, we were able to clearly understand that the variable for Final Fasting Plasma Glucose (FFPG) had the absolute highest correlation with a positive diagnosis for diabetes, clearly indicating that there is a drastic increase in average values for diabetic patients as opposed to non-diabetic patients. This is similar to our visualization of Fasting Plasma Glucose (FPG). Additionally, we were able to visualize lipid profiles such as HDL and LDL cholesterol, indicating that there is a significant distribution change that is correlated to the onset of the disease.

4. Train Test Split

The dataset was split into training and testing sets using a 70:30 ratio, where 70% of the dataset is training and 30% for testing. A stratified sampling approach was implemented based on the target variable. This is to ensure that the distribution remains consistent amongst both the training and testing datasets. There are 3012 observations in the training set and 1291 observations in the test set. The class distribution was well preserved with approximately 69.7% non-diabetic cases and the rest to be diabetic cases in both sets.

By maintaining a consistent class distribution, it prevents bias during model training and ensures that the model is evaluated on a representative sample of the data. The approach will help to improve the reliability and generalisability of the model's performance.

5. One Hot Encoding

Categorical variables which are smoking and drinking were encoded using one-hot encoding to prepare them for modelling. The OneHotEncoder with handle_unknown="ignore" was used to ensure consistency when handling unseen categories.

For variables that are already binary such as Gender, family_history were not encoded as they are already in a suitable numerical format. The encoded variables were then concatenated with the numerical features. The final dataset has 21 features in total.

6. Machine Learning Model

a) Feature Importance (Random Forest)

To look beyond simple visual correlations, we utilized the Random Forest classifier to extract the Feature Importance of our variables. Because Random Forest is an advanced ensemble method made up of multiple decision trees, it can automatically capture complex, non-linear relationships and calculate exactly how much each feature contributes to reducing classification error.

Our initial hypotheses have been validated through the feature importance analysis, which revealed that biochemical markers, especially Fasting Plasma Glucose (FPG/FFPG), are the dominant, most statistically significant factor for the onset of Diabetes. This was followed closely by Age and Cholesterol Levels (HDL/LDL), which proves that demographic factors significantly compound an individual's risk. Although lifestyle factors like smoking and drinking habits have been included, the algorithm has revealed that direct physiological measurements are of the most importance. This, from a healthcare perspective, is a highly actionable result, as it reveals which factors physicians need to focus on during initial patient screenings to prevent false negatives.

b) Standard Scaler

Before feeding our data into our machine learning algorithms, we used the StandardScaler from SciKit-Learn to process our final set of features.

This data preprocessing step was absolutely necessary because our data set includes numerical data with drastically different units and magnitudes (Age ranges from 22 to 93, while FPG ranges from 1.78 to 6.99). Any machine learning algorithm that relies on distance calculations (like K-Nearest Neighbors) or margin optimization (like Support Vector Classifiers and Logistic Regression) is very sensitive to unscaled data. If unscaled data is used, features with larger numbers would overwhelm the algorithm, causing an inaccurate prediction. By scaling our data so that all features have a mean of 0 and a standard deviation of 1, we ensured that all health indicators would contribute equally to our model's learning process, thus optimizing our model's prediction accuracy for all our algorithms.

c) Logistic Regression

The Logistic Regression model was used as the standard classification model to predict diabetes in a clear way. The model achieved an accuracy of 0.9078, precision of 0.8886 and a recall of 0.7954, this means it correctly identified about 80% of diabetic cases. This metric is very important in healthcare especially, as missing a true case can have serious consequences. The F1-score of 0.8394 shows a strong balance between precision and recall. As well, the model had very fast training and prediction times, making it efficient for practical use. Overall, Logistic Regression provided strong, balanced, and interpretable performance.

d) Decision Tree

The Decision Tree model was used in order to address non-linear relationships through rule-based splits. The model achieved an accuracy of 0.8737 and a precision of 0.7850, which means it had a higher amount of false positives in comparison to the other models. It did however, achieve a slightly higher recall of 0.8031, meaning it identified more diabetic cases than Logistic Regression. The model achieved a 0.7939 F-1 score and this reflects

the trade-off between precision and recall. Even though the model is easy to interpret and is able to capture complex patterns, it is more prone to overfitting, which can make it less reliable, especially on new data.

e) K-Nearest Neighbours (KNN)

The KNN model was trained using the encoded dataset after standardizing all features with StandardScaler, since KNN is distance-based and sensitive to feature scale. After testing k values from 1 to 20, the best result was found at k = 9, with an accuracy of 0.8916.

The final KNN model achieved 0.8916 accuracy, 0.9061 precision, 0.7161 recall, and 0.8000 F1-score. While KNN performed well overall and was especially strong in precision, its recall was lower than that of stronger models, indicating it missed more actual diabetes cases. Given the project's priority of identifying diabetic individuals, KNN was not selected as the final model, as other models achieved a better balance between recall and precision.

f) Support Vector Classifier (SVC)

The SVC model was trained on the scaled and encoded dataset using the default RBF kernel. It achieved 0.9101 accuracy, 0.9009 precision, 0.7903 recall, and 0.8420 F1-score, making it one of the strongest models in the project.

Compared with KNN, SVC performed better overall, especially in recall and F1-score, showing a better balance between identifying diabetes cases and limiting false positives. Although SVC performed very well, it was slower and more computationally intensive than Logistic Regression. Since Logistic Regression produced results very similar to those of SVC while being simpler, faster, and easier to interpret, SVC was not chosen as the final model.

Final Model and Fine-Tuning

After comparing the performance of all models, Logistic Regression was selected as the final model for further tuning. Although SVC achieved slightly higher accuracy and F1-score, Logistic Regression was chosen because it still performed strongly, had competitive recall, was much faster, and was easier to interpret. This was especially important for this project, since understanding how the predictors relate to diabetes risk adds value beyond prediction alone.

To fine-tune the chosen model, GridSearchCV with 5-fold cross-validation was used. The tuning process tested several values of the regularization parameter CCC, including 0.01, 0.1, 1, 10, and 100, while keeping the penalty as L2 and the solver as lbfgs. The model was also tested with and without class weighting. The best parameter combination returned by the grid search was C=0.1, with an L2 penalty, the LBFGS solver, and no class weighting.

Using these settings, the tuned Logistic Regression model achieved an accuracy of 0.9078, a precision of 0.8908, a recall of 0.7928, and an F1-score of 0.8390 on the test set. The confusion matrix showed that the model correctly classified 862 non-diabetic individuals and 310 diabetic individuals, while incorrectly classifying 38 non-diabetic individuals as diabetic and missing 81 diabetic individuals.

When the tuned model was compared with the original Logistic Regression model, the results were almost identical. The baseline model had the same accuracy of 0.9078, along with a precision of 0.8886, a recall of 0.7954, and an F1-score of 0.8394. Although the tuned model slightly improved precision, both recall and F1-score decreased slightly. Because

recall is especially important in this project, since missing a diabetes case is more serious than incorrectly flagging a non-diabetic individual, the tuning process did not lead to a meaningful improvement in overall performance.

Overall, the fine-tuning results suggest that the original Logistic Regression model was already performing close to its best on this dataset. Rather than substantially improving the model, the tuning process mainly confirmed that Logistic Regression was a stable and reliable final choice.

References

https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html
https://www.kaggle.com/datasets/pkdarabi/diabetes-dataset-with-18-features
https://www.who.int/news-room/fact-sheets/detail/diabetes

Appendix

Figure 1: Boxplot of All Numerical Variables

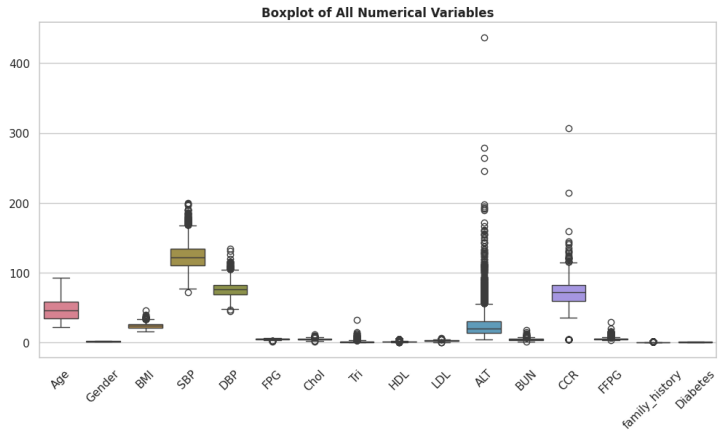

Figure 2: Model Results

	Model	Accuracy	Precision	Recall	F1	Train Time	Predict Time
0	Logistic Regression	0.907823	0.888571	0.795396	0.839406	0.097761	0.000623
1	Decision Tree	0.873741	0.785000	0.803069	0.793932	0.146608	0.001059
2	K-Nearest Neighbors (KNN)	0.891557	0.906149	0.716113	0.800000	0.006253	0.122163
3	SVC	0.910147	0.900875	0.790281	0.841962	0.361165	0.165758

Figure 3: Logistic Regression Results

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Logistic Regression Fine Tuned
Accuracy: 0.9078
Confusion Matrix (rows=actual, cols=pred):
[[862  38]
 [ 81 310]]

                  precision    recall  f1-score   support

No Diabetes (0)       0.91      0.96      0.94       900
   Diabetes (1)       0.89      0.79      0.84       391

       accuracy                           0.91      1291
      macro avg       0.90      0.88      0.89      1291
   weighted avg       0.91      0.91      0.91      1291

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Github Repository Link: https://github.com/kohyuqing09/OMIS3020_Project_Group1